

CS 4114

Operating System

Introduction and Architecture

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What is an Operating System

An operating system (OS) is the foundational software that manages all of a device's hardware and software resources. It acts as an essential bridge between the user, applications, and physical components (like the CPU and memory), allowing your device to function smoothly.  Wikipedia +1

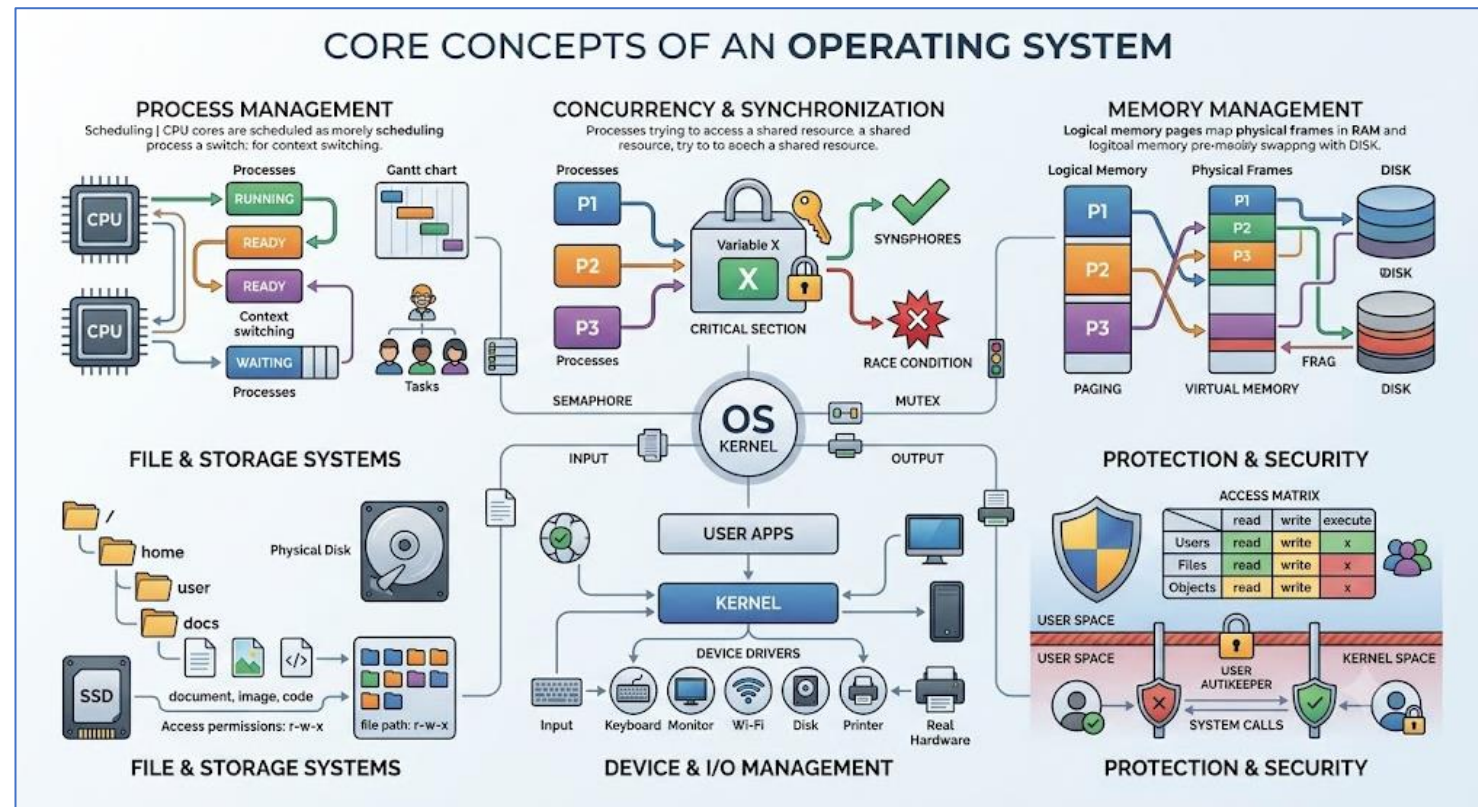
Core Functions

An operating system handles the complex, low-level tasks required to run a computing device:  TechTarget

- **Memory Management:** Keeps track of storage and allocates memory space for various programs to ensure they don't interfere with one another.  Britannica
- **Processor (CPU) Allocation:** Decides which processes run, when they run, and for how long to keep the device running efficiently.  Britannica

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- **File Management:** Organizes data logically on storage devices (like a hard drive or cloud) and controls read/write access. [GeeksforGeeks +2](#)
- **Device Control:** Provides a standardized way for applications to communicate with hardware peripherals, like a mouse, keyboard, or printer. [Lenovo](#)





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Popular Operating Systems

Different devices require specialized operating systems tailored to their hardware and interface designs:  Wikipedia

- **Desktop & Laptop Computers:** Dominated by [Microsoft Windows](#), Apple's macOS, and various distributions of [Linux](#).
- **Mobile Devices:** Largely powered by Google Android and Apple's [iOS](#).
- **Servers & Enterprise:** Linux and Unix variants are the industry standard for powering massive web servers and cloud infrastructure.  Wikipedia +1



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When it comes to desktop PCs and laptops, **Microsoft Windows** remains the dominant leader globally, though Apple's ecosystem (macOS/OS X) and Linux maintain substantial, highly dedicated user bases.

According to global data from tracking platforms like *Statcounter*, the operating system market share breakdown for desktop and laptop computers is structured as follows:

Global Desktop OS Market Share



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Operating System	Approximate Global Share	Key Target Segment & Usage
Microsoft Windows	62% – 72%	General consumers, corporate environments, enterprise systems, and gaming.
Apple macOS / OS X	14% – 15%	Creative professionals, software developers, and consumers within the Apple ecosystem.
Linux (Desktop)	3% – 4%	Software engineers, IT administrators, servers, and privacy-focused power users.
ChromeOS	1.5% – 2%	Education sectors (K-12 schools), lightweight budget laptops (Chromebooks).
<i>Other / Unknown</i>	~10% - 18%	Highly customized, secure, or un-tracked proprietary systems.



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Internal Version Breakdown (Windows & Mac)

1. The Windows Ecosystem

Within the Windows ecosystem, a massive multi-year migration has been taking place. Following Windows 10 approaching its official end-of-life status, **Windows 11** has firmly established itself as the majority leader:

- **Windows 11:** ~71.7%
- **Windows 10:** ~26.4%
- **Older Versions (Win 7, 8, XP):** ~2% combined

2. The Apple Ecosystem

Analytics web tracking typically splits Apple desktop computers into two historical categories based on user-agent reporting:

- **OS X (Older naming convention / Legacy legacy versions):** ~10%
- **macOS (Modern operating system versions):** ~4.5%



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Regional Differences

OS preferences change drastically depending on where you look:

- North America & United States:** Apple has a much stronger foothold here. Windows sits lower at roughly **64%**, while Apple's desktop ecosystem claims over **21%** of the market. ChromeOS also peaks here at around **4.7%** due to its heavy integration into US school districts.
- South America:** Windows holds an ironclad grip with nearly **89%** of the total desktop market share, leaving macOS with under 6%.
- Asia:** Windows commands roughly **74%**, while Linux and ChromeOS make up minor shares.
- (Note: In highly specialized fields like supercomputing globally, **Linux** actually controls 100% of the market).*



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While Linux has a relatively small (though growing) slice of the standard desktop and laptop market, it is actually the most widely run operating system in the world. Instead of running on consumer PCs, Linux completely dominates behind the scenes, powering global infrastructure, mobile phones, and the cloud.

Linux runs almost exclusively in the following key areas:

1. Cloud Computing and Web Servers

The modern internet operates almost entirely on Linux.

- **Web Hosting:** Over **96% of the top one million web servers** run on Linux distributions (like Ubuntu, Debian, or Red Hat).
- **The Cloud:** Roughly **90% of all public cloud infrastructure** (including Amazon Web Services (AWS), Google Cloud Platform, and Microsoft Azure) operates on top of Linux.



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2. Smart Devices & Mobile (Android)

If you own an Android device, you are running Linux.

- Android Smartphones:** Android uses a modified version of the **Linux kernel** at its core. Because Android commands roughly 72% of the global mobile market, Linux is technically the most installed end-user OS on earth.

3. Supercomputers (100% Dominance)

- Every single one of the **top 500 fastest supercomputers** in the world runs a specialized distribution of Linux. No other operating system is used in high-performance scientific computing because Linux can be highly customized and stripped of unnecessary visual bloat.



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4. Embedded Systems & Smart Devices

Because Linux is open-source and modular, manufacturers can shrink it down to fit on tiny microchips. It powers:

- Smart Home Devices:** Routers, smart TVs (like Samsung's Tizen and LG's webOS), and smart appliances.
- Automotive:** The infotainment and autonomous driving systems for brands like Tesla, Toyota, and Mercedes-Benz.
- Aerospace:** Spacecraft infrastructure, including SpaceX's Falcon 9 rockets and NASA's Mars rovers



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5. Software Development & Cybersecurity

- DevOps & Programming:** A vast majority of software engineers write code on or deploy software to Linux environments.
- Ethical Hacking:** Cybersecurity professionals utilize specific distributions like *Kali Linux*, which comes pre-bundled with hundreds of tools for digital forensics and penetration testing.

6. Gaming (The Portable Breakthrough)

- While Linux historically struggled with video games, Valve's **Steam Deck** (which runs on a modified version of Arch Linux called SteamOS) has turned Linux into a massive handheld gaming platform.



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5 types of operating systems

While you might have heard Microsoft Windows or macOS mentioned in conversation, you can find an extensive list of operating systems built around different priorities and professional needs. Five overarching categories of operating systems you can explore include the following.

Batch

When executing programs, batch operating systems do so in batches. Because devices with batch operating systems submit tasks to an operator, this type of system is ideal for time-consuming tasks with large files. This type of system is widely used for payroll, data entry, and similar operations.

While beneficial for large tasks, you might have difficulty debugging batch operating systems when errors arise, and implementing a strong batch operating system can be resource-intensive.



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Time-sharing

Time-sharing operating systems switch between multiple tasks, each with a time limit. This type of system benefits users from multiple locations by enabling them to use it simultaneously and providing quick responses to user input.

Distributed

Distributed operating systems connect multiple machines together through a shared network. Each computer has its own memory and operating system but can communicate with other devices through the distributed system. This allows data sharing to happen quickly for system users, limits data processing delays, and reduces the load placed on the host machine.

Telecommunications networks and airline reservation controls use this type of system because it allows for fast data exchange, is resilient against system failures, and can scale easily.



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Network

Network operating systems connect computers and other devices through a shared network, known as a [local area network \(LAN\)](#). Through this network, you can access the same documents and share files and physical devices. Network operating systems offer excellent stability and remote access, and support common file management, security, user administration, and system maintenance.

Real-time

Real-time operating systems perform tasks close to real-time, adhering to strict timing requirements. This type of operating system is best when precision is highly important, making it the operating system of choice for air traffic control systems, weapon control systems, robots, and medical imaging.

**END OF PRESENTATION
THANK YOU**